

WILLIAM ROHREN

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EDUCATION

Texas A&M University - GPA: 3.15

BSME: Bachelor of Science in Mechanical Engineering

College Station, TX

Graduation: Dec 2026

SKILLS

Simulation & Analysis

Ansys FEA, Femap/Nastran, Topology Optimization

Test & Instrumentation

Strain Gauges, Bridge Circuits, DAQ, Instron (MEEN 361)

Design & Fabrication

SolidWorks, CNC Machining, 3D Printing, Plasma Cutting

Electronics & Embedded Systems

Altium, KiCAD, Mixed-Signal PCB, SMT, Embedded C

EXPERIENCE

Texas A&M SAE Aero Design Team

College Station, TX

Product Owner, Structures (Officer)

Sep 2026 – Present

- Co-lead a 9-engineer structures subteam, owning all structural design for the 2026-27 competition airframe
- Delegate tasks, hold the build schedule, and review subteam analysis for proper modeling assumptions

Structures & Materials Science (SMS) Engineer

May 2025 – Sep 2026

- Owned the design of a cantilevered nose beam motor mount, consolidating propulsion electronics in the fuselage and simplifying the motor wiring harness
- Swept Ansys topology optimization from 5–50% mass retention, then reworked the 20% output into a buildable 9.1 oz balsa truss, FEA-verified under 25 lbf landing and 12 lbf per-motor thrust loads
- Modeled the main wing's 1/64 in leading-edge skin three ways (3D solids, 2D shells via midsurfacing, omitted) to set FEA fidelity; all agreed within 5%, so the team dropped skin from the model and cut mesh size and solve time
- Designed 5 nose structures across 3 airframe iterations, all flight tested, with no structural failures under design load conditions

Rapid Prototyping Studio (RPS)

College Station, TX

Student Technician

Jan 2024 – Present

- Trained 500+ students on shop equipment and safety across FDM/SLA printing, laser cutting, and manual machining
- Maintained and repaired studio fabrication equipment, managing \$14,000 in annual consumables and repairs
- Self-trained on CNC plasma cutting to bring an unused Torchmate table into service; authored the shop's 34-page SOP and trained 8 staff on the machine

PROJECTS

Senior Capstone: Spacecraft Lifting Fixture (MGSE) - Southwest Research Institute

Aug 2025 – May 2026

- Owned strain instrumentation and FEA correlation for a 6-person team designing a modular ISO 7 cleanroom lifting fixture rated for 200–1000 kg spacecraft
- Built a custom DAQ reading eight 350 Ω gauges in half-bridge pairs through four HX711 24-bit ADCs on an Arduino Nano, bonding gauges and fabricating harnesses on site
- Measured 76.2 and 71.8 $\mu\epsilon$ against an Ansys prediction of 76.5 $\mu\epsilon$ (0.4% and 6.2%) during a 1300 lb proof lift, using SolidWorks split lines to extract simulated strain at the exact gauge locations

Project SAVIOR - 16-Channel Strain Data Recorder

Apr 2026 – Present

- Designing a 16-channel strain DAQ to measure in-flight wing loads against FEA predictions (eight ADS1220 24-bit ADCs, Teensy 4.1, 6-DoF IMU, ± 200 g accelerometer); schematic drafted, layout, fab, and testing pending
- Selected bridge topology per channel by what each measurement must reject: half-bridge type II for bending, 3-wire series-pair arms for axial

Custom Atmega32U4 Development Board

Aug 2024 – Jan 2025

- Designed a custom ATmega32U4 dev board in Altium, programmed bare-metal in C, manufactured via JLCPCB
- Debugged USB-C implementation failures across two board revisions; successfully flashed firmware via ICSP

ADDITIONAL INTERESTS

Astro-Photography, Fishing, Trumpet, PC-Building, 3D Printing, Embedded Systems, RC Aircraft